

# Root Decay and Fire Affect Soil Pipe Formation and Morphology in Forested Hillslopes with Restrictive Horizons

**Ian N. Leslie**

**Robert Heinse\***

Soil & Land Resources Division  
Dep. of Plant, Soil, & Entomological Sci.  
Univ. of Idaho  
Moscow, ID 83844-2339

**Alistair M. S. Smith**

Dep. of Forest, Range and Fire Sci.  
Univ. of Idaho  
Moscow, ID 83844-1136

**Paul A. McDaniel**

Soil & Land Resources Soil &  
Land Resources Division  
Dep. of Plant, Soil, & Entomological Sci.  
Univ. of Idaho  
Moscow, ID 83844-2339

Soil pipes or large horizontal macropores are hillslope–hydrologic features affecting critical processes such as subsurface drainage, streamflow, erosion, hillslope stability, and discharge response to rain fall and snow melt. However, information on soil pipe formation, morphology, and distribution on forested hillslopes is very limited, impeding the incorporation of these important preferential flow pathways into mechanistic hydrologic models. The objectives of this study were to (i) quantify the spatial distribution and physical characteristics of soil pipes, (ii) investigate soil pipe origin, and (iii) examine the role of fire in pipe formation. We excavated three 6.12- by 6.12-m plots down to a water-restricting horizon within a 2-ha portion of a steep forest hillslope in northern Idaho. Excavation followed consecutive timber harvest and prescribed burning operations. Detailed documentation of soil pipe diameter, direction, gradient, and connectivity from these excavated soil profiles were used to derive morphologic attributes and distribution of soil pipes. The majority of soil pipes were attributed to decayed tree roots with a median depth of 0.32 m corresponding to the beginning of the restrictive soil horizon. Median path length for branched and nonbranched soil pipes was 2.21 m and 0.89 m, respectively with a median tortuosity of 1.23. Our findings indicate that soil pipe formation and distribution within this hillslope was influenced by tree root growth, decay, slope gradient, and root limiting soil horizons. Fire and the combustion of roots and tree stumps was the primary mechanism that connected soil pipes to the surface. The large frequency and extent of soil pipes documented in this study suggests that estimations of water transport without consideration for these macropores, while useful, are likely underestimated.

**Abbreviations:** UIEF, University of Idaho Experimental Forest.

Preferential pathways and connected macropores commonly described as soil pipes have been shown to play an important role in hillslope hydrology affecting subsurface drainage (Tromp-van Meerveld and McDonnell, 2006), streamflow (Uchida et al., 1999), erosion (Wilson, 2011), hillslope stability (Uchida et al., 2001), discharge response to snow melt (Roberge and Plamondon, 1987), and precipitation (Uchida et al., 2005). While some larger-scale investigations of preferential flow pathways including soil pipes in coniferous forested hillslopes have occurred (Anderson et al., 2009a), detailed investigations of soil pipe morphology and distribution within coniferous hillslopes of the United States are very limited despite their prevalence in some forest ecosystems (e.g., Fig. 1). Here we used detailed excavation of forested field plots to document the spatial distribution and hydrologically important morphological characteristics of soil

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\*Corresponding author (rheinse@uidaho.edu).

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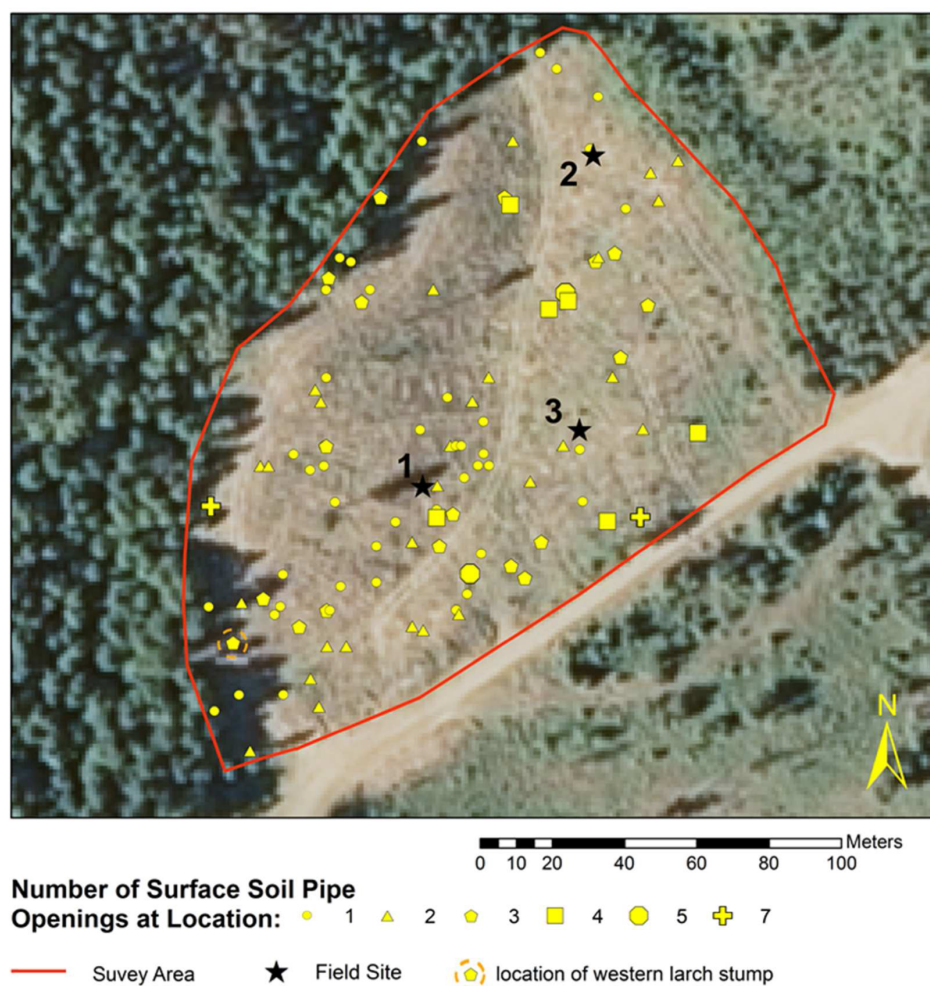
pipes forming in soils characterized by a water restrictive soil horizon. We made use of common management practices in managed forests to examine the impact of fire on soil pipe formation. Hydrological models may critically underestimate preferential flow pathways in these hillslopes without a better understanding of soil pipe morphological characteristics, development, and potential rapid formation in managed forests via combustion.

A critical component of understanding how soil pipes influence hillslope hydrology is the quantification of morphological characteristics such as diameter, tortuosity (a measure of straightness), and connectivity (the connection of macropores in topological space), which have been suggested to control subsurface flow velocity and maximum drainage capacity (Beven and Germann, 1982; Anderson et al., 2009b; Kitahara, 1993; Uchida et al., 2005). However, as Uchida et al. (2001) noted in their seminal review on soil pipes, only a few detailed studies on soil pipe morphology and distribution in forest soils have been conducted. Of these, Aubertin (1971) employed excavations to

inventory soil pipe-macropore morphology in mixed deciduous forest soils in Ohio; however, important characteristics such as tortuosity and connectivity were not reported. Noguchi et al. (1997, 1999), Kitahara (1989, 1994), and Terajima et al. (2000) documented soil pipe lengths, diameters, and tortuosities on coniferous forested hillslopes in Japan. Neither Noguchi et al. (1999) nor Kitahara (1989, 1994) published tortuosity for branched soil pipes. Terajima et al. (2000) and Uchida et al. (1999) did describe branched soil pipes, but did not investigate the full extent of the branches. These studies provide useful but limited knowledge of soil pipe formation, distribution, and connectivity in forest soils, which critically impairs the ability to incorporate the effect of soil pipes into hydrological models of forested watersheds (Nieber and Sidle, 2010; Uchida et al., 2005). Additionally, soil pipe formation in coniferous forests may be more complex than previous studies suggest as post-fire field observations suggest soil pipes may rapidly form or may be partially enhanced through the combustion of decaying tree roots. Ferreira et al. (1997) proposed that burned out

root channels observed in a catchment dominated by conifers [cluster pine (*Pinus pinaster* Aiton)] were partially responsible for low amounts of overland flow directly following a forest fire when compared to ripped-plowed areas. Moody et al. (2013) noted that the presence of post-wildfire surface depressions from burned out stumps and roots may increase pipe flow and surface water storage. However, we are not aware of any previously published study investigating the role of fire in soil pipe formation.

Soil pipes forming within forested watersheds with water-restricting soil horizons or bedrock may be particularly important given the potential for soil pipe development resulting from timber harvesting practices, which leave behind evenly-aged decaying tree root systems coupled with subsequent wild fires, prescribed burns, and perched water table formation. Throughout large forested areas of northern Idaho, seasonally perched water tables commonly form over restrictive horizons such as fragipans and similar argillic horizons (McDaniel et al., 2001, 2008). In this study, we added to the limited knowledge of soil pipe morphology and distribution needed to better model hillslope hydrology by measuring soil pipe diameter, direction, gradient, tortuosity, and connectivity from ex-



**Fig. 1.** Location of Field Plots 1, 2, and 3 on the a 2-ha clearcut study area at the Flat Creek Unit of the University of Idaho Experimental Forest approximately 11 km north of Troy, ID. Soil pipes are prevalent in this area, with 204 surveyed soil pipe surface openings across this 2-ha clearcut documented after a prescribed burn. In this exploratory survey, we documented the location, largest diameter, and the maximum linear length as measured from the surface opening. For the survey of pipe openings at the soil surface, soil pipes were defined as having a diameter of at least 0.03 m and extending at least 0.10 m from the opening in any linear direction. Soil pipes were recorded as occurring at the same location if within 0.5 m of each other.

cavated soil profiles on steep forest hillslopes in northern Idaho underlain by a water-restricting horizon. We also evaluated soil pipe origin and investigated the role of fire in soil pipe formation.

## MATERIALS AND METHODS

### Study Area

Our field site was on a 2-ha clearcut within a portion of a small drainage located in the Flat Creek Unit of the University of Idaho Experimental Forest (UIEF) approximately 11 km north of Troy, ID at 46°50'29.01" N lat, 116°46'29.65" W long, and at 940 m asl (Fig. 1). The UIEF ranges in daily mean temperature from 0 to 17°C from winter to summer, respectively, and receives approximately 690 mm of annual precipitation (Sandoval et al., 2007). Over 60% of annual precipitation in the region occurs as light rain and snow from November to April (Brooks et al., 2004).

Timber species before harvesting consisted of a diverse conifer forest composed of grand fir [*Abies grandis* (Douglas ex D. Don) Lindl.], western redcedar (*Thuja plicata* Donn ex D. Don), Douglas-fir [*Pseudotsuga menziesii* (Mirb.) Franco], ponderosa pine (*Pinus ponderosa* P. Lawson & C. Lawson), and western larch (*Larix occidentalis* Nutt.). Timber was last harvested the summer of 2009 using a cut-to-length harvester-forwarder method. Fir, larch, and pine were previously harvested in the area from 1927 to 1929. The study site underwent a prescribed burn using a strip-firing technique on 14 Oct. 2010.

## Soil Description

The soils are mapped as the Santa series and are ashy silt loams on 3 to 35% slopes classified as coarse-silty, mixed, super-active, frigid Vitrandic Fragixeralfs (Soil Survey Staff, 2004; Soil Survey Division, 2003). A soil profile description was obtained from an excavation 1 m to the side of Plot 1 (see Fig. 1) at the same hillslope position while soil profile descriptions for Plots 2 and 3 were obtained during the excavation of the plots. Table 1 provides selected soil morphological and physical characteristics from the field plots. The O horizon is composed of slightly to moderately decomposed conifer needles, bark, and grasses and ranges in depth from 0 to 4 cm. A thin layer of relatively unweathered volcanic ash (C horizon) from the 1980 Mount St. Helens eruption is found at the base of the O horizon and ranges in thickness from 0.5 to 1 cm. The O and C horizons are not uniformly present across the plots, likely having been disturbed during the timber harvesting operations. The A horizon for these soils are thin and very similar to the Bw horizon with the exception of many more very fine and fine roots from grasses and forbs and breaking to granular structure. Each soil profile exhibited Bw horizons between 8 and 30 cm in depth underlain by a transitional BE horizon extending to a depth of up to 46 cm. The E horizon ranged from 45 to 65 cm in depth. All soil profiles contain a water-restricting Btxb (fragipan) horizon beginning between 51 and 65 cm in depth and running roughly parallel to the slope of the soil surface. The fragipan horizons at the

**Table 1. Summary of selected field plot soil morphological and physical characteristics.**

| Horizon | Depth<br>cm | Dry color                        | Texture         | Structure†               | Rupture resistance (dry) |
|---------|-------------|----------------------------------|-----------------|--------------------------|--------------------------|
| Plot 1  |             |                                  |                 |                          |                          |
| O       | 0–4         | NA                               | NA              | NA                       | NA                       |
| C       | 4–4.5       | 2.5YR 7/1                        | NA              | NA                       | NA                       |
| A       | 4.5–10      | very pale brown (10YR 7/3)       | silt loam       | 2 m sbk parting to 1 f g | slightly hard            |
| Bw      | 10–28       | very pale brown (10YR 7/3)       | silt loam       | 2 m sbk                  | slightly hard            |
| BE      | 28–46       | pale brown (10YR 6/3)            | silt loam       | 1 f sbk                  | slightly hard            |
| E       | 46–65       | very pale brown (10YR 7/3)       | silt loam       | massive                  | slightly hard            |
| Btxb1   | 65–90       | very pale brown (10YR 7/3)       | silt loam       | 2 c sbk                  | very hard                |
| Btxb2   | 90–102+     | light yellowish brown (10YR 6/4) | silty clay loam | 2 c abk                  | very hard                |
| Plot 2  |             |                                  |                 |                          |                          |
| O       | 0–2.5       | NA                               | NA              | NA                       | NA                       |
| C       | 2.5–3.5     | light gray (2.5YR 7/1)           | NA              | NA                       | NA                       |
| A       | 3.5–10      | brown (7.5YR 4/3)                | silt loam       | 1 m sbk parting to 1 f g | slightly hard            |
| Bw1     | 10–23       | brown (7.5YR 4/3)                | silt loam       | 1 m sbk                  | slightly hard            |
| Bw2     | 23–37       | brown (7.5YR 5/3)                | silt loam       | 1 m sbk                  | slightly hard            |
| BE      | 37–46       | brown (10YR 5/3)                 | silt loam       | 1 f sbk                  | slightly hard            |
| E       | 46–55       | very pale brown (10YR 7/3)       | silt loam       | massive                  | slightly hard            |
| Btxb    | 55–60+      | yellowish brown (10YR 6/4)       | silt loam       | 1 f sbk                  | very hard                |
| Plot 3  |             |                                  |                 |                          |                          |
| O       | 0–4         | NA                               | NA              | NA                       | NA                       |
| C       | 4–5         | light gray (2.5YR 7/1)           | NA              | NA                       | NA                       |
| A       | 5–8         | brown (7.5YR 4/3)                | silt loam       | 1 m sbk parting to 1 f g | slightly hard            |
| Bw      | 8–30        | pale brown (10YR 6/3)            | silt loam       | 2 m sbk                  | slightly hard            |
| BE      | 30–45       | pale brown (10YR 6/3)            | silt loam       | 1 f sbk                  | slightly hard            |
| E       | 45–51       | very pale brown (10YR 7/3)       | silt loam       | massive                  | slightly hard            |
| Btxb    | 51–62 +     | yellowish brown (10YR 6/4)       | silt loam       | 1 f sbk                  | very hard                |

† 1 = weak; 2 = moderate; f = fine; m = medium; c = coarse; g = granular; sbk = subangular blocky; abk = angular blocky; NA = not applicable.





Fig. 2. Example of an excavated soil profile at Plot 1 (decimeter-scale reference).

clearcut are believed to be part of a Pleistocene paleosol and are common throughout the Palouse region of eastern Washington and northern Idaho (McDaniel et al., 2001, 2008). The coarse to very coarse prismatic structure typically observed with Btxb horizons in these soils was not observed at Plots 2 and 3; however, the Btxb horizon was restrictive to roots, dense, brittle, and had very hard rupture resistance.

### Excavation and Soil Pipe Morphological Characterization

Soil excavations were performed on three 6.12- by 6.12-m field plots centered as much as possible on a single decaying stump (Fig. 1). Field plots were selected to facilitate previous geophysical surveys (electrical resistivity tomography) with the aim of indirectly characterizing soil pipes forming from stump and root combustion during the prescribed burn reported in Leslie and Heinse (2013).

Plots were on 33, 27, and 31% slopes, for Plots 1, 2, and 3, respectively. Plot 1 was on a northeast-facing slope and Plots 2 and 3 were on northwest facing slopes. Plots were excavated

along 25 parallel profiles in 0.255-m increments perpendicular to the slope during August and September 2011. Each of the 25 profiles (see, for example, Fig. 2) was excavated by hand to a minimum depth of 0.50 m using sharpshooter shovels. The excavation soil profiles were roughly perpendicular to the gradient of the slope. A perpendicular orientation of soil pipes to the excavation profile was defined as  $0^\circ$ , with positive gradients indicating an upslope orientation and negative gradients indicating a downslope orientation (Fig. 3). To enhance interpretability of profiles, profile faces were carefully prepared with trowels to expose soil structure and soil pipes. Soil pipes on the face of the profiles were classified as soil pipes if they had an opening diameter of  $\geq 0.01$  m and extended linearly 0.10 m or more into the soil profile in any direction or gradient.

Spatial locations ( $x,y,z$ ) of soil pipes on profiles were recorded along with the vertical and horizontal diameters, gradients, directions of the soil pipe (Fig. 3), and evidence of their origin was noted. Relative  $z$ -locations of the soil pipes and the excavation area topography were determined using a water level attached to a precision pressure transducer (SMS tensiometer,

Soil Measurement Systems, Tucson, AZ). Spatial locations were recorded to within  $\pm 0.01$  m from their center, soil pipe diameters were recorded within  $\pm 0.005$  m, and direction and gradient were recorded to within  $\pm 5^\circ$ . Some directional bias may be imparted into the measurements from the perpendicular orientation of the soil profiles to the slope gradient. However, this placement transects the direction and gradient in which lateral flow would be expected and is, thus, of most interest to hill-slope hydrology. Vertical and horizontal soil pipe diameters measured from plot excavations were averaged to produce a mean soil pipe diameter.

The connectivity of soil pipes was traced using polyethylene tubing with an outside diameter of 6 mm. The tubing was flexible enough to negotiate turns  $< 90^\circ$  and fit within soil pipes as long as the pipe was not blocked with soil, undecayed root, or

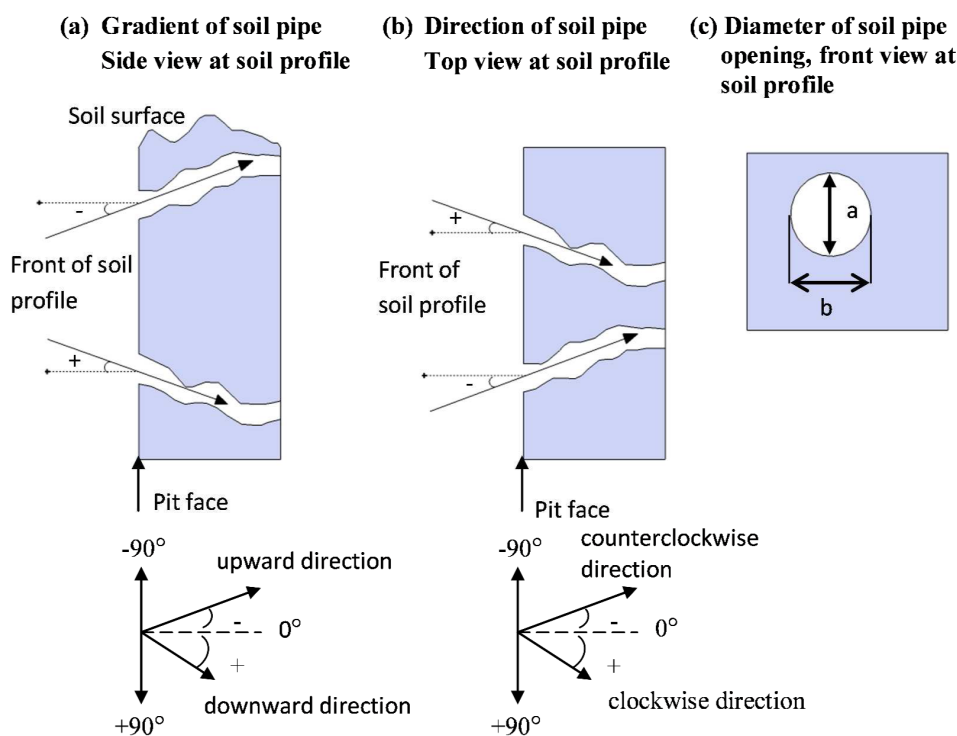


Fig. 3. Soil pipe cross-sections direction, gradient, and diameter measurements, where “a” and “b” are the vertical and horizontal diameters, respectively (adapted from Noguchi et al., 1997, Fig. 2).



other obstructions. We acknowledge that soil pipes with obstructions may still provide preferential flow pathways; however, this method provided a rapid means of documenting the connectivity of mostly empty soil pipes. The tortuosity ( $T$ ) was determined following the methods of Noguchi et al. (1999):

$$T = \sum l_i / L \quad [1]$$

$$\sum_{i=1}^{n-1} l_i = \sum_{i=1}^{n-1} \left[ (x_i - x_{i+1})^2 + (y_i - y_{i+1})^2 + (z_i - z_{i+1})^2 \right]^{0.5} \quad [2]$$

$$L = \left[ (x_e - x_b)^2 + (y_e - y_b)^2 + (z_e - z_b)^2 \right]^{0.5} \quad [3]$$

where  $x$ ,  $y$ , and  $z$  are the Cartesian coordinates for beginning and ending locations of connected soil pipes (nodes),  $l_i$  is the estimated path length of the  $i$ th soil pipe as measured from soil profile to soil profile,  $L$  is the linear distance, and  $n$  is the number of profiles ( $\geq 3$ ) the soil pipe continued through. Length of soil pipes from excavated field plots was calculated using Eq. [3] for branched and nonbranched soil pipes extending through three or more excavation soil profiles. Tortuosity for soil pipe networks was calculated along pathways connecting terminal nodes where a node represents a soil pipe at the face of an excavated soil profile, and a terminal node is a node that only connects to one other node. Soil pipe networks were mapped using the connectivity information recorded during the field plot excavations. Three-dimensional spline-interpolated visualizations of soil pipes were produced using ParaView (2013).

## RESULTS

### Soil Pipe Formation

Before excavation of the field plots, a survey of soil pipe openings at the soil surface across the 2-ha clearcut documented 204 soil pipes (Fig. 1). Soil pipe surface opening diameters ranged from 3 to 44 cm, with the linear lengths ranging from 10 to more than 100 cm; lengths beyond 100 cm could not be measured outside of the field plots before excavation. The median for soil pipe surface opening diameter was 10 cm with a standard deviation of 6.2 cm. The median linear length was 47 cm with a standard deviation of 22 cm. Out of 99 locations where soil pipe surface openings were measured, 43% of locations occurred as individual soil pipes, and 57% of locations had two to seven soil pipes (Fig. 4). At locations where more than one soil pipe was documented, all soil pipe openings formed from the combustion of dead tree stumps and roots. This was clearly evident from depressions left in the soil from the combustion of the stump and the location of the soil pipe openings on the sides of the depressions, beneath the surrounding ground level. For example, two soil pipe openings were generated after the prescribed burn at Plot 1 from the decaying stump and roots (Fig. 5a and 5b). The charred remains of root material were observable in the surrounding soil pipe walls (Fig. 5c). Excavation of plot 1 revealed that Soil Pipe Opening 1C (Fig. 5c) connected to the nonbranched Soil Pipe 1C with a pipe length of 1.17 m

and tortuosity of 1.03 (Fig. 6) and exhibited evidence of burning through two soil profiles (0.51 m). In situ burned root bark was also observed in the branched Soil Pipe 1A ~1 m down slope of the soil pipe's surface opening as shown in Fig. 5c, near the remains of the stump at Plot 1. Segments of Soil Pipes 1C and 1A down slope of the above-mentioned soil profile sections were attributed to tree root decay.

Also documented in the survey of soil pipe openings at the soil surface was the rapid formation of soil pipes from the combustion of a western larch stump (Fig. 1) and subsequent formation of soil pipes as it continued to smolder over a 7-d period (Fig. 5d and 5e). Soil pipes formed from the larch stump ranged from 25 to 56 cm in length and 12 to 15 cm in diameter at the surface opening.

From field plot excavations, we attributed initial soil pipe formation (i.e., origin) to two mechanisms: tree root decay and rodent burrows. We further subdivided the soil pipes formed from decayed roots into two categories: soil pipes filled with <25% of decaying root matter and those filled with >25% decaying root matter. Soil pipes with <25% decaying root matter were primarily free of organic matter, debris, or soil, but some had thin sections of bark or organic stains lining the channel. About 96% of all soil pipes showed evidence of formation from decaying tree roots, with more than 86% containing <25% decaying root matter. Only 5% and 8% of soil pipes, at field Plots 2 and 3 respectively, exhibited evidence of originating from rodent activity. No soil pipes at Plot 1 were attributed to formation from rodent activity.

### Soil Pipe Morphological Characteristics

A summary of soil pipe origin, gradient, direction, and diameter from these three plots are presented in Table 2. Mean soil pipe diameters ranged from 1.00 to 32.50 cm, with a mean of 4.3 cm. The largest mean soil pipe diameters in Table 2 are associated with larger roots of the decaying stump. However the majority of soil pipes had much smaller diameters as represented by the median soil pipe diameters, which were very similar for each plot, within about 1 cm. Analysis of soil pipe depth showed more than 85% of the soil pipes

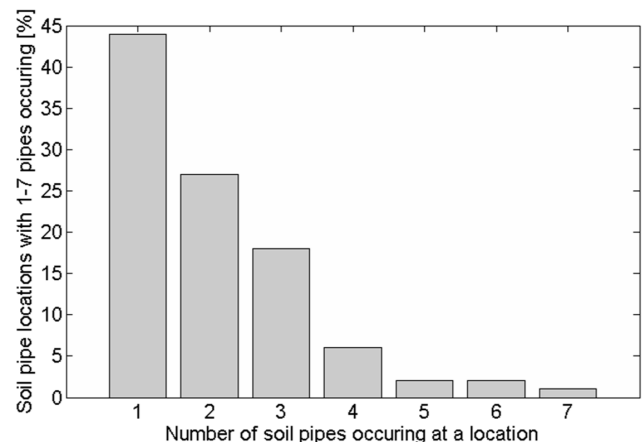
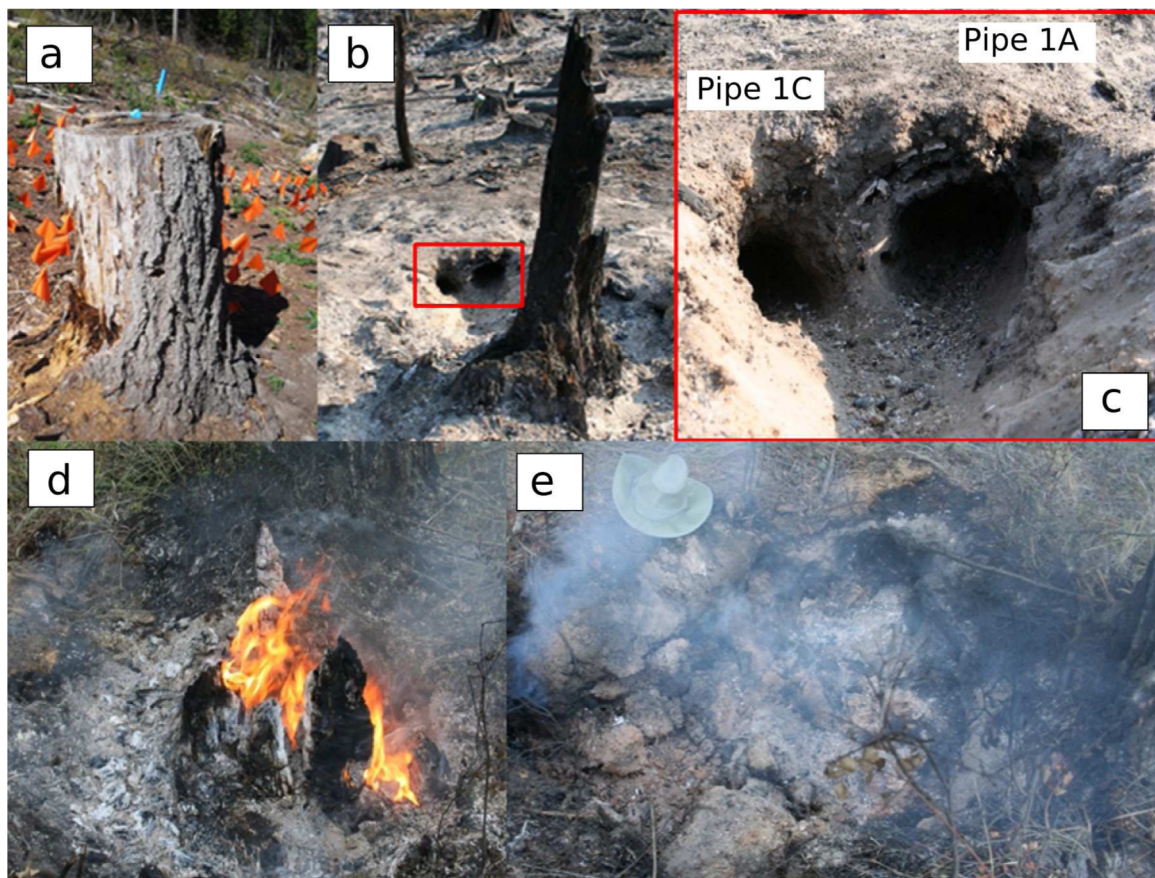


Fig. 4. Percentage of soil pipes locations with one to seven soil pipe surface openings occurring at that location.



**Fig. 5.** Decaying stump at Plot 1 (a) pre and (b) post-prescribed burn; (c) close up of Soil Pipes 1C and 1A; (d) burning larch stump on 14 Oct. 2010, the day of the prescribed burn; (e) burning roots from larch stump on 21 Oct. 2010.

occurred within the extent of the O, A, Bw, and BE horizons (~0 to 46 cm) and 15% occurred below this depth within the Btxb horizon. The median depth for all soil pipes was 32 cm with 35% of the soil pipes between depths 30 and 40 cm.

**Table 2. Summary of soil pipe morphological characteristics;  $n$  = number of soil pipes documented from excavated soil profiles.**

| Field plot                                   | 1                | 2                | 3                |
|----------------------------------------------|------------------|------------------|------------------|
| $n$                                          | 306              | 244              | 229              |
| Soil pipes > 25% decayed root remains (%)    | 9                | 9                | 3                |
| Soil pipes < 25% decayed root remains (%)    | 91               | 86               | 89               |
| Soil pipes from rodent activity (%)          | 0                | 5                | 8                |
| Median, mean and mode of gradient (degrees)  | -20, -16, -25    | -20, -17, -20    | -20, -18, -20    |
| Minimum of gradient (degrees)                | -90              | -80              | -60              |
| Maximum of gradient (degrees)                | 80               | 90               | 90               |
| Median, mean and mode of direction (degrees) | 10, 4, 20        | 10, 3, 10        | 0, -6, 5         |
| Minimum of direction (degrees)               | -90              | -85              | -90              |
| Maximum of direction (degrees)               | 90               | 85               | 70               |
| Oblique positive gradients (%)               | 12               | 12               | 8                |
| Oblique negative gradients (%)               | 88               | 88               | 92               |
| Oblique positive direction (%)               | 58               | 56               | 50               |
| Oblique negative direction (%)               | 42               | 44               | 50               |
| Median, mean and mode of diameter (cm)       | 2.93, 3.83, 2.00 | 4.00, 5.00, 3.00 | 3.00, 4.05, 2.00 |
| Standard deviation of diameter               | 3.68             | 3.28             | 3.47             |
| Minimum mean diameter (cm) †                 | 1.00             | 1.00             | 1.00             |
| Maximum mean diameter (cm)                   | 32.50            | 23.00            | 27.50            |

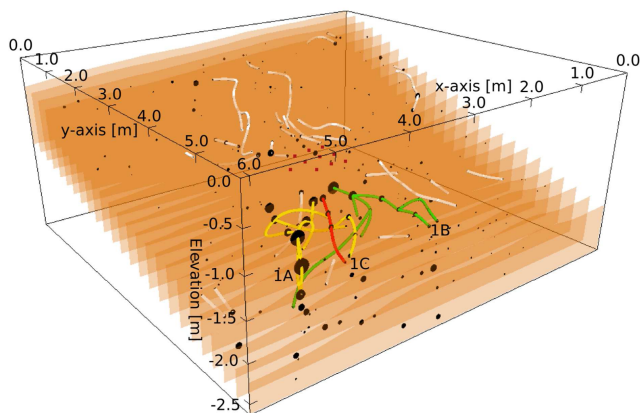
† Minimum and maximum mean diameter: the minimum–maximum of the average of the horizontal and vertical diameter measurements of a soil pipe as illustrated in Fig. 3.

The median gradient of soil pipes for each plot was approximately  $-20^\circ$ , close to the median slope gradient of  $-17^\circ$  for Plots 1, 2, and 3 (Table 2). The median direction of soil pipes was similar between plots, within  $10^\circ$  (Table 2). While oblique

positive and negative directions were nearly equal over the three excavation plots, more than 88% of soil pipes had oblique negative gradients (Table 2). Seventy-five, 75, and 81% of soil pipes from Plots 1, 2, and 3 had directions between  $-50^\circ$  and  $50^\circ$ .

Soil pipe locations and connectivity documented from excavation profiles are illustrated in Fig. 6, 7, and 8. Approximately 54% of all soil pipes were more than 25 cm in length, with 44% having lengths greater than 0.5 m. Seven branched soil pipes were documented from the three field plots (Fig. 6: 1A, 1B; Fig. 7: 2A–2D; Fig. 8: 3A). The soil pipe path length for all soil pipes with  $\geq 3$  nodes had a median of 1.36 m ranging from 0.51 to 5.07 m. Median path length for branched soil pipes was 2.21 m with a standard deviation of  $\pm 1.01$  m compared to





**Fig. 6.** Spline-interpolated visualization of soil pipes with more than two nodes at Plot 1 (ParaView 2013). Individual soil pipe openings are shown as black circles scaled to mean soil pipe diameters. Red squares represent decayed stump location. Highlighted Soil Pipes 1A–1C are referenced in the text.

nonbranched soil pipes with a median path length of 0.89 m and standard deviation of  $\pm 0.55$  m. Tortuosities for nonbranched and branched soil pipes are given in Table 3. Tortuosity of soil pipes continuous through three or more excavation profiles was low, with a medium tortuosity of 1.23.

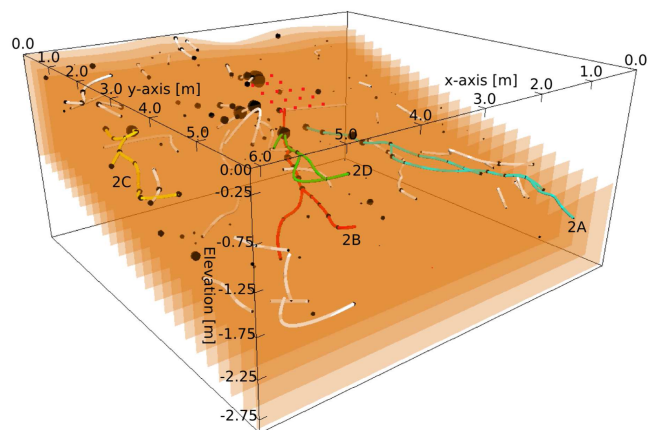
## DISCUSSION

### Soil Pipe Formation and Morphology

Our investigation on the role of fire in soil pipe formation suggests that while tree root decay is a key factor in soil pipe initiation, fire is an important factor in soil pipe enhancement through the opening of pipes to the soil surface and the decrease of root material within the soil pipe through combustion. Analysis of UIEF records indicated that major harvest operations at this site last occurred from 1927 to 1929 indicating that the majority of the soil pipes present formed from tree mortality originating from the 1927 to 1929 harvest and natural mortality in subsequent years. Based on their observations, Beven and Germann (1982) suggested that macropores formed from the decay of tree roots could last at least 50 to 100 yr and that soil pipes open to the surface could be capable of channeling surface runoff from precipitation and melting snow. Ferreira (1996) recorded infiltration rates directly over burned out root channels of up to  $250 \text{ mm h}^{-1}$  on recently burned hydrophobic conifer forests plots. This compares to infiltration rates of  $1.8 \text{ mm h}^{-1}$  on plots without such channels as reported by Shakesby et al. (2000).

Saturated soil conditions and saturation-excess runoff has been documented to occur for up to 7 mo out of the year in these and similar soils where the upper boundary of the water restricting fragipan is within 40–100 cm of the surface (McDaniel et al., 2008). Prescribed burns or forest fires over areas with large tree mortality could rapidly change the hydrology in these areas from the opening of these potential preferential flow pathways to the surface.

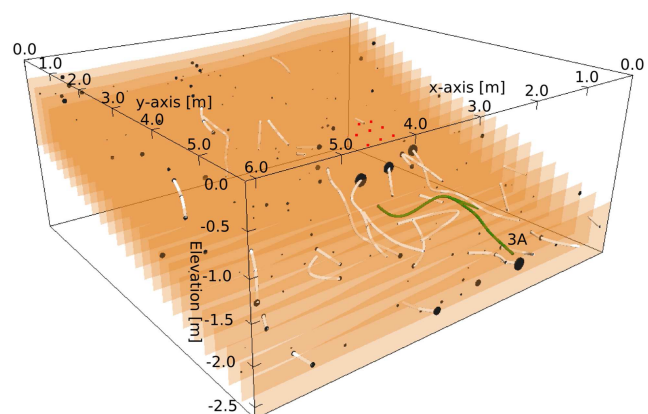
The facilitation of surface drainage by the opening of soil pipes to the surface through combustion may require a mini-



**Fig. 7.** Spline-interpolated visualization of soil pipes with more than two nodes at Plot 2 (ParaView 2013). Individual soil pipe openings are shown as black circles scaled to mean soil pipe diameters. Red squares represent decayed stump location. Highlighted Soil Pipes 2A–2D are referenced in the text.

mum pipe diameter. Based on simulated soil pipes in a silt loam, Wilson (2011) suggested that soil pipes originating from biological origins such as animal burrows or decayed tree roots require initial diameters greater than 4 mm, or clogging from internal erosion would be nearly instantaneous from water directly entering the pipe under a constant head. The influence of soil pipe clogging from internal erosion and resulting rapid pore water pressure increases within soil pipes has been examined through numerical modeling (Wilson and Fox, 2013) and through in situ experiments (Midgley et al., 2013). While all soil pipes documented in this study were larger than 4 mm, the influence of additional sediment or ash from surface erosion into soil pipes and potential clogging has not yet been examined. Further study on the potential for soil pipe clogging from surface sediment or ash is needed to better evaluate the effect on hillslope drainage from the combustion driven opening of soil pipes to the surface.

Soil pipe diameter and angle also play a role in determining soil pipe contributions to hillslope hydrology. After comparing several studies on soil pipe flow, Uchida et al. (2005) con-



**Fig. 8.** Spline-interpolated visualization of soil pipes with more than two nodes at Plot 3 (ParaView 2013). Individual soil pipe openings are shown as black circles scaled to mean soil pipe diameters. Red squares represent decayed stump location. Highlighted Soil Pipe 3A referenced in the text.



**Table 3. Statistical summary of path length ( $l$ ) and tortuosity ( $T$ ) of nonbranched and branched soil pipes;  $n$  = number of soil pipes  $\geq$  three nodes.**

|                        | plot | n  | Path length |      |      |        | Tortuosity |       |      |        |
|------------------------|------|----|-------------|------|------|--------|------------|-------|------|--------|
|                        |      |    | Min         | Max  | Mean | Median | Min        | Max   | Mean | Median |
| m                      |      |    |             |      |      |        |            |       |      |        |
| Branched soil pipes    | 1    | 2  | 0.63        | 3.97 | 2.27 | 2.19   | 1.05       | 26.91 | 3.26 | 2.17   |
|                        | 2    | 4  | 0.95        | 5.07 | 2.77 | 2.43   | 1.07       | 5.68  | 2.08 | 1.52   |
|                        | 3    | 1  | 1.07        | 1.96 | 1.51 | 1.50   | 1.14       | 1.70  | 1.37 | 1.27   |
| Nonbranched soil pipes | 1    | 58 | 0.51        | 2.69 | 1.11 | 1.00   | 1.00       | 1.27  | 1.07 | 1.04   |
|                        | 2    | 16 | 0.36        | 2.18 | 0.99 | 0.91   | 1.00       | 3.39  | 1.30 | 1.09   |
|                        | 3    | 20 | 0.52        | 2.41 | 1.05 | 0.84   | 1.00       | 3.99  | 1.30 | 1.07   |

cluded that the maximum capacity of pipe flow and the rainfall intensity at which maximum capacity is reached are dependent on soil pipe diameter and angle (i.e., gradient and direction). Comparing soil pipe diameters to other studies is very difficult since soil depth, texture, fauna, flora, parent material, slope gradient, and climate likely influence soil pipe development. In addition, few detailed studies examining soil pipe and macropore morphology in forested hillslopes are available (Terajima et al., 2000; Uchida et al., 2005). However, it is interesting to note that the range in soil pipe diameters from our plots (1.0–32.5 cm) are similar to the range in soil pipe diameters documented from the forested Hakyuchi Experimental Watershed, Japan (1.0–20 cm). Both the Hakyuchi and the U.I. Flat Creek plots are on steep coniferous forested slopes and underlain by soil horizons approximately 0.50 m in depth that are more water-restricting than the overlying horizons (Uchida et al., 2005). While more studies are needed, this tentatively suggests that soil pipe diameter and angle could be indirectly estimated through vegetation, topography, and soil type.

Tree root growth and decay appear to be the dominate mechanisms for initial soil pipe formation. This is indicated by approximately 96% of all soil pipe cross-sections initially forming from tree root decay. This is in contrast to the 37% of macropores attributed to subsurface erosion and root decay, respectively, reported by Noguchi et al. (1997) for macropore-soil pipes occurring in conifer-dominated hillslopes in Japan. Considering the large percentage of soil pipes with <25% decaying root matter or other debris, we suggest water movement through the soil pipes as one possible factor in preventing the filling of these macropore pathways with debris as suggested by Tsukamoto et al. (1988). Physical and numerical modeling simulations of pipe flow by Wilson (2011) and by Wilson and Fox (2013) suggest that soil pipes clogged due to internal erosion are capable of rapidly unclogging and continuing to flow; however, organic debris was not present or included in these simulations. Water flow through soil pipes in this study is also suggested by the longer lengths of soil pipes observed at our field plots. Noguchi et al. (1999) attributed greater soil pipe length and connectivity of soil pipes forming in forested hillslopes in Japan to the extent and duration of elevated water table levels, suggesting that the frequency of saturated subsurface flow was an important factor in maintaining or developing longer soil pipes. All plots excavated in this study contained soil pipes with path lengths in excess of 2 m. Soil Pipe Networks

2A and 2B (Fig. 8) at Plot 2 had path lengths ( $l$ ) of 5.07 and 3.75 m, respectively. Additionally, soil morphological evidence strongly suggests the presence of perched water tables at the field plots. McDaniel and Falen (1994) reported a strong correlation between E horizon thickness and extent of redoximorphic features with perched water table height. Few to common medium Fe masses within peds were observed in the BE horizon of Plots 1 and 3 and within the E horizons at all plots. The presence of E horizons above the fragipan and redox concentrations within the BE soil horizon at our field plots indicate perched water tables occur within these hillslopes for as long as 7 mo during the year based on nearby monitoring studies (McDaniel et al., 2001; Brooks et al., 2004; McDaniel et al., 2008). The presence of perched water tables, lack of debris with the pipes at the time of excavation, and morphological features conducive to pipe flow suggest that these soil pipes conduct water during some portion of the year.

Rodent activity appears to play a small yet potentially important role in soil pipe network formation at our plots. While it was not possible to link soil pipes formed from rodent activity to a specific species, the northern pocket gopher (*Thomomys talpoides*) likely accounts for nearly all of the soil pipes classified as originating from rodent activity. This is supported by the lack of northern pocket gopher activity at Plot 1, where none of the soil pipes exhibited evidence of originating from rodent activity. Despite comprising a small percentage of total soil pipes, rodent activity may play an important role in producing connected soil pipe networks. Rodent burrows connect otherwise unconnected soil pipes formed from tree root decay to produce a branched soil pipe network (2C in Fig. 7) at Plot 2 with a path length ( $l$ ) of 2.4 m.

Previously reported tortuosities of soil pipes in forest soils were close to 1, that is, almost linear pathways. Kitahara (1989) estimated tortuosities between 1.0 and 1.1 from casts of plaster filled soil pipes, and Noguchi et al. (1999) reported forest soil macropore tortuosity ranging from 1.00 to 1.52 with a mean tortuosity of 1.14. In a later study, Kitahara (1994) reported a mean tortuosity of 1.03 for soil pipes excavated from a 0.092 by 0.092 m<sup>2</sup> by 0.3 m long trench. In our study, tortuosity for nonbranched soil pipes ranged between 1.00 and 3.99 (Table 3), a much larger range than reported by Kitahara (1989, 1994) and Noguchi et al. (1999). However, the median tortuosity of nonbranched soil pipes from each plot was very close to 1 (Table 3) as was the overall median tortuosity of 1.23 for all soil pipes.

The low tortuosity of nonbranched soil pipes may be partially explained from elongations of tree roots in the up and down slope directions and narrow distribution in direction. However, low tortuosities would also be expected for soil pipes undergoing internal erosion via preferential flow. The maximum measured tortuosity, 27, for branched soil pipes is due to a branch of the Soil Pipe 1B, which almost loops to connect with an adjacent branch connected to the same soil pipe (Table 3; Fig. 6); however, since the two adjacent branches of this soil pipe are connected up slope, the high tortuosity would not be expected to restrict water flow in the down slope direction. This highlights the importance of obtaining detailed information on soil pipe connectivity for hydrological modeling.

### Influence of Slope Gradient and Soil Morphology

Slope gradient and soil morphology appear to have strongly influenced soil pipe gradient, direction, and distribution with depth within these hillslopes. The increase in the frequency of soil pipes with depth up to between 30 and 40 cm and median depth of soil pipes at 32 cm corresponds to the beginning of the BE horizon. High bulk densities typical of the E and Btxb horizons ( $>1.6 \text{ Mg m}^{-3}$ ; Reuter et al., 1998) just below the BE horizon limits root growth within these horizons.

The gradient and direction of soil pipes on these hillslopes also suggests hillslope gradient plays an important role in determining soil pipe gradient and direction. Chiatante et al. (2003) concluded that roots on steep slopes tend to develop a bilateral fan-shape in which the lateral roots are asymmetrically distributed in the upslope and downslope orientations due to preferential elongations of lateral roots in those orientations. A bilateral fan-shape could partially account for the 75 to 81% distribution in soil pipe opening direction within  $-50$  and  $50^\circ$ . The tendency for lateral root growth in the upslope and downslope orientations could also explain the dominance of the oblique negative gradient distribution observed for soil pipe. While not directly comparable, it is interesting to note that the majority of macropores (including soil pipes) in A and B horizons on steep coniferous forested slopes hillslopes reported by Noguchi et al. (1997) also had directions between  $-50$  and  $50^\circ$ . Additionally, Noguchi et al. (1997) attributed 55 to 70% of macropores in the A and B horizons to root channels.

### SUMMARY AND CONCLUSIONS

Root channels provide the primary initial mechanism for soil macropore-pipe formation in these forested hillslopes, which has important implications for modeling preferential flow pathways in these soils. Soil pipe gradient, direction, and depth suggested that the slope gradient and soil morphology, particularly the restriction of the Btxb horizon to root penetration, influence the distribution of soil pipes occurring from root channels. Additionally soil pipe gradient, tortuosity, and the presence of long connected soil pipes suggest preferential flow through these soil pipes could facilitate high flow velocities. The large density of soil pipe surface openings occurring throughout the clearcut

attributed to the combustion of decaying tree roots suggests portions of a hillslope hydrology may rapidly change over very brief periods from the occurrence of fire on the landscape. This could occur from the rapid opening of numerous soil pipes to the surfaces that are readily available to conduct water from the surface and through portions of a hillslope. The presence of well-connected and extensive soil pipes within the field plots formed primarily from tree root decay alone indicates that soil pipes appear relatively stable over time periods required for tree root decay and would typically experience perched water tables. We demonstrated that fire is an important factor in soil pipe formation, one that has been overlooked in the literature until now.

Our findings support previous suggestions that macropore pathways may account for the saturation-threshold driven mean pore velocity and lateral saturated hydraulic conductivities reported for previously forested hillslopes in the Palouse region (Brooks et al., 2004; McDaniel et al., 2008). However, our findings suggest preferential flow pathways are not limited to the Ap horizon in these soils and that soil pipe distribution and morphology in these soils are dependent primarily on tree root growth, which is in turn influenced by limiting soil horizons and slope gradient. Our data suggests hydrological models that take into account tree root growth, decay, and fire interactions may be able to bridge the gap between field plot and hillslope measurement scales by better accounting for hydraulic variability from macropore preferential flow pathways not accounted for in small-scale measurements.

Additionally, our findings suggest that soil pipes in these soils may play an important hydrological role in these hillslopes given the increase in soil pipe frequency above the BE horizon, the presence of perched water tables within these soils above the Btx horizon, and soil pipe morphological characteristics that would allow these pipes to function as preferential pathways. The number and extent of soil pipes documented in this study strongly suggests that estimations of permeability, conductivity, and solute transport made based only on soil textural information, while useful, are likely underestimated.

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